



FT-4x00 Datasheet

MEMS Force Sensor

Features

- Excellent Sensitivity, Linearity and Reliability
- Fast Response
- Excellent Survival Force
- Excellent Electrical and Thermal Stability
- Differential Output
- Ultra-Low Power
- Ultra-Small 5-pin WLCSP

Description

The FT-4x00 series MEMS force sensor represents the highest performance, smallest size and most reliable force sensing solution in the market. Designed to serve a broad range of applications such as buttons, sliders and screens. The sensor can be used with or without capacitive touch controllers to enable a robust 3D touch experience.

The sensing solution enables new gapless industrial designs, provides 10x higher reliability compared to mechanical switches and can be used with any material including metal, glass, plastic, ceramic, wood, leather, etc.

The sensor along with proprietary algorithms accurately measure force associated with a touch event on any surface. This MEMS-based solution performs precisely and consistently over temperature and humidity changes and is immune to radio frequencies (RF) and electromagnetic interference (EMI).

The sensor is compatible with standard surface-mount assembly processes, including pick-and-place equipment and reflow soldering profiles, facilitating high-volume manufacturing on both rigid PCB and flexible FPCB substrates.

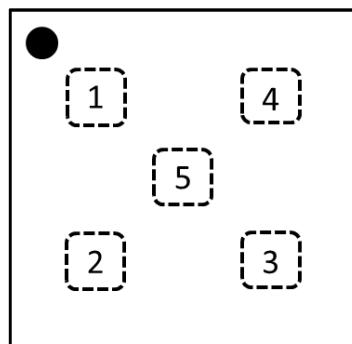
Applications

- Automotive
- Industrial
- Consumer: Smartphones, Tablets, Wearables
- IoT: Smart Home Devices
- Input Devices: Stylus, Touchpads
- Gaming and AR/VR: Force-Sensing Controls

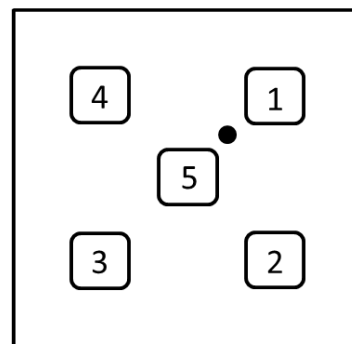
Characteristics

- Supply Voltage: 1.62 V to 5.5 V
- Force Range: 0 N to 10 N
- Absolute Maximum Force: 125N
- Package Size: 1.33 x 1.33 x 0.57 mm³

Pin Configuration



Top View



Bottom View

Pin Description

SOLDER BUMP	PIN NAME	PIN DESCRIPTION
1	SPOS	Sensor Positive Output Terminal.
2	SE	Sensor Negative Excitation Voltage.
3	SNEG	Sensor Negative Output Terminal.
4	VCC	Sensor Positive Excitation Voltage. Connect to stable low-noise power supply (e.g. LDO source or DRV from AFE) for the best performance.
5	MECH	Mechanical Stability. This pin is used for mechanical stability only. The MECH pin must be soldered to the board and not connected electrically to any net.

Ordering Information

PART NUMBER	OPERATING TEMPERATURE	GRADE	DESCRIPTION
FT-4100A00	-20°C to +70°C	Consumer	3000pc 7-inch reel
FT-4100A00SR	-20°C to +70°C	Consumer	300pc sample reel
FT-4300B00	-40°C to +85°C	Industrial/Medical	3000pc 7-inch reel
FT-4300B00SR	-40°C to +85°C	Industrial/Medical	300pc sample reel
FT-4500B00	-40°C to +85°C	Automotive Grade 3	3000pc 7-inch reel
FT-4500B00SR	-40°C to +85°C	Automotive Grade 3	300pc sample reel
FT-4500C00	-40°C to +105°C	Automotive Grade 2	3000pc 7-inch reel
FT-4500C00SR	-40°C to +105°C	Automotive Grade 2	300pc sample reel
FT-4100DK	N/A	N/A	Full Design Kit (FT-4100 + AF-3068)
FT-4100PCK	N/A	N/A	Design Kit, Sensor Module only (FT-4100 + AF-3068)

Absolute Maximum Ratings

Stresses beyond those listed may cause permanent damage to the device. These are stress ratings only; functional operation of the device under these or any other conditions beyond those indicated in the operational sections is not implied. Prolonged exposure to absolute maximum rating conditions may degrade device reliability.

PARAMETER	VALUE
VCC to SE	-0.3 V to +6 V
Applied Force (flat probe tip with 0.4 mm thick 70A rubber, T _A = +25 °C) ¹	125 N
Operating Temperature Range	See Ordering Information
Storage Temperature	-65 °C to +150 °C
Lead Temperature (Soldering 10 s)	+260 °C
Shock Absorption	±10,000 g
Electrostatic Discharge Protection (ESD)	2000 V (HBM), 750 V (CDM)

¹ This is survival limit, not for continuous operation.

Electrical Characteristics

VCC¹ = 1.80 V ± 10%, SE = 0 V, T_A = -40 °C to +85 °C, typical values are at T_A = +25 °C, min/max limits 100% tested at room temperature, unless otherwise noted.

When pair sensor with AF-3x68, please note that DRV = 1.62 V.

PARAMETER	SYMBOL	CONDITIONS	MIN.	TYP.	MAX.	UNITS
Supply Current	I _{CC}	TA = 100% duty cycle		240	295	μA
		+25 °C 1% duty cycle		2.4		
Off-state Current	I _{OFF}	VCC = SE		0		μA
Zero Force Offset	ZFO	TA = +25°C	-9		18.2	mV
Sensitivity ²		TA = +25°C	4.44	6.21	8.1	mV/N
		TA = -40°C to +85°C	3.31		9.23	
Temperature Coefficient of Sensitivity ²	TCS			-9.66		μV/N/°C
Full-Scale Range	FSR		8	10		N
Bridge Resistance	R _{BR}			7.50		kΩ
Linearity ²		TA = +25 °C, best straight-line fit	0.99		1.00	
Hysteresis		TA = +25°C		1	3.5	%FSR
Noise	N			7.88		nV/√Hz
Common-Mode Voltage	V _{CM}			V _{CC} /2		V

¹ Use linear interpolation from 1.8 V to calculate performance values for applications using different VCC values.

² Values based on characterization data over temperature.

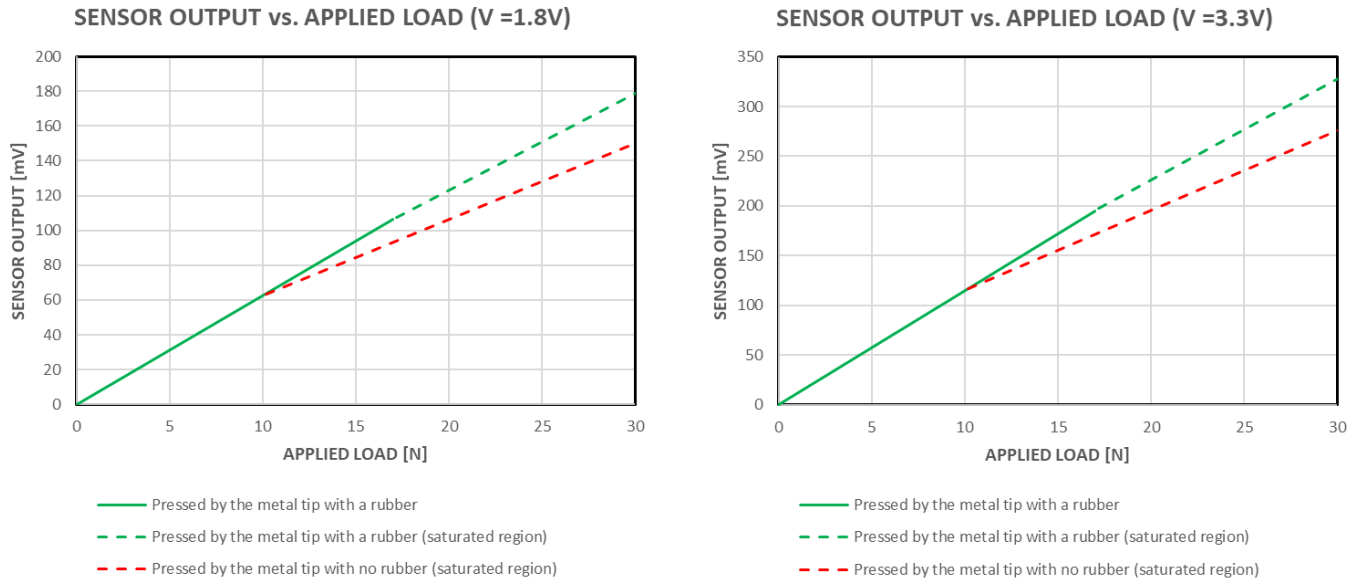
VCC¹ = 3.30 V ± 10%, SE = 0 V, T_A = -40 °C to +85 °C, typical values are at T_A = +25 °C, min/max limits 100% tested at room temperature, unless otherwise noted.

PARAMETER	SYMBOL	CONDITIONS	MIN.	TYP.	MAX.	UNITS
Supply Current	I _{CC}	TA = 100% duty cycle		440	541	μA
		+25 °C 1% duty cycle		4.4		
Off-state Current	I _{OFF}	VCC = SE		0		μA
Zero Force Offset ²	ZFO	TA = +25°C	-16.5		33.37	mV
Sensitivity ²		TA = +25°C	8.14	11.39	14.85	mV/N
		TA = -40°C to +85°C	6.07		16.92	
Temperature Coefficient of Sensitivity ²	TCS			-17.71		μV/N/°C
Full-Scale Range	FSR		8	10		N
Bridge Resistance	R _{BR}			7.50		kΩ
Linearity ²		TA = +25 °C, best straight-line fit	0.99		1.00	
Hysteresis		TA = +25°C		1	3.5	%FSR
Noise	N			7.88		nV/√Hz
Common-Mode Voltage	V _{CM}			V _{CC} /2		V

¹ Use linear interpolation from 3.3 V to calculate performance values for applications using different VCC values.

² Values based on characterization data over temperature.

Typical Operating Characteristics



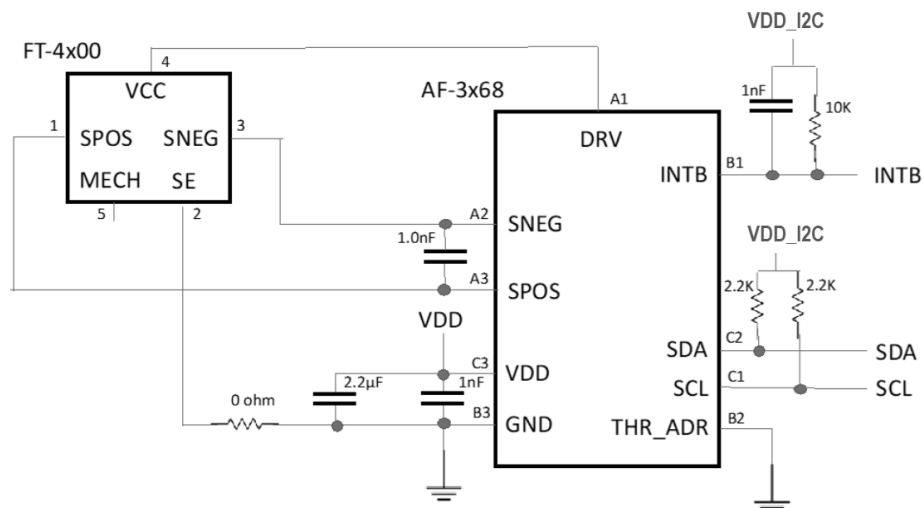
Application Information

This sensor is a compact silicon device designed to measure applied forces from 0 N to 10 N with milli-newton-level resolution. The achievable resolution depends on the bit depth and performance of the external analog-to-digital converter (ADC) used in the system.

Individual sensor or sensor arrays may be mounted beneath a touch surface to enable force-based human-machine interfaces (HMI). When appropriately integrated, the sensors support a range of use cases including discrete force-sensitive buttons, trigger mechanisms, stylus detection, and single-axis robotic or control inputs.

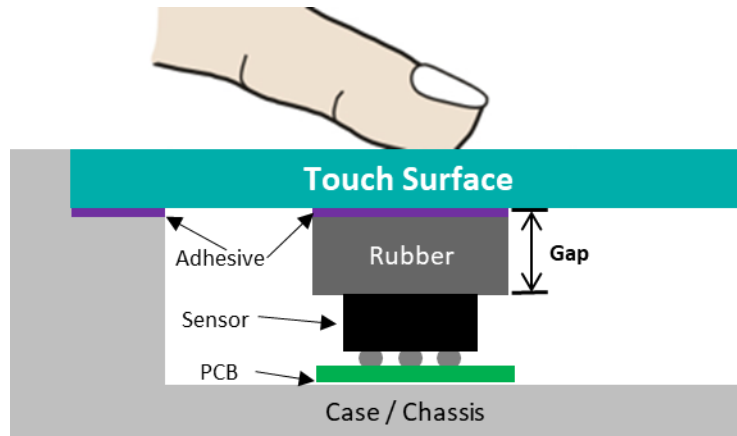
Typical Operating Circuit

Figure below demonstrates a simple setup utilizing Semtech sensing solution to communicate with an I²C host: FT-4x00 Force Sensor + AF-3x68 (Ultra-low Power Analog Front-End with I²C Interface).



System Stack-up

In button-type implementations, the sensor is actuated through a layer of silicone rubber positioned beneath the device's external surface (made by any material). The rubber layer protects the sensor and helps absorb manufacturing tolerances commonly encountered in display, button, or button-under-surface applications. With a properly designed stack-up, the sensor can reliably detect applied forces from a few grams up to 1000 grams or more. For mechanical integration, a 2 mil (50 μm) double-sided adhesive is recommended for bonding the silicone rubber to the underlying stack-up.



Recommended Sensor Stack-up With Spacer (Requires Contact on Both Sides)

A 70 Shore A adhesive-backed silicone rubber—for example, Ames Rubber Manufacturing 0.015 SIL GRY—is recommended to accommodate typical vertical tolerances. To accommodate a mechanical tolerance of $\pm 100\text{ }\mu\text{m}$, a 400 μm -thick rubber layer is used, providing a nominal compression of approximately 110 μm during actuation (see below table). To absorb larger mechanical tolerances, contact Semtech for further guidance.

MECHANICAL TOLERANCES	GAP	RUBBER THICKNESS	RUBBER COMPRESSION	
			MIN.	MAX.
$\pm 100\text{ }\mu\text{m}$	290 μm	400 μm	2.5%	60%

Rubber Design for Absorbing Mechanical Tolerances

The size of the rubber element should be larger than the sensor's footprint (1.33 mm x 1.33 mm); a 2 mm diameter or 2 mm x 2 mm square pad is generally sufficient.

MEMS Force Sensor Analog Front-End (AFE) References

Below table shows the AFE with I2C interface which allows the sensor to be connected to a host. For a list of recommended microcontroller reference designs contact Semtech.

VENDOR	PART NUMBER	PACKAGE
Semtech	AF-3x68*	WLCSP

* Semtech's MEMS force sensor solution can work with any host

Electrical Design, Layout, Placement and Handling

Electrical Design Considerations

The sensor is a sensitive analog device. All routing, bypassing, and component placement practices described in this section are intended to preserve signal integrity, minimize noise coupling, and ensure reliable operation across the operating temperature and supply-voltage range.

- Power all analog circuitry from a low-noise supply source, preferably a dedicated on-board LDO.
- Match the SPOS and SNEG trace lengths and minimize parasitic asymmetry between the sensor and the AFE differential inputs.
- At AFE side, two bypass capacitors are required, 2.2 μF & 1 nF, to reduce noise on the power supply. Use higher value bypass capacitors for noisy power supplies. Ensure that both capacitors are placed close to the AFE VDD pin.
- Connect sensor SE pin via a 0-ohm resistor directly to the grounded terminal of the bypass capacitors next to the AFE. The 0-ohm resistor and the bypass capacitors should be ≤ 3 mm away from the AFE.
- Add a 1 nF differential filter capacitor (highlighted in the table below) between SPOS and SNEG and place the differential capacitor close to the differential inputs of the AFE.

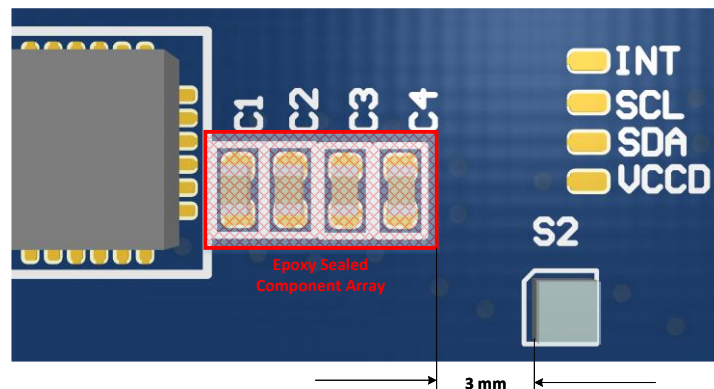
The differential capacitor value affects both sensor power-on settling time and the analog noise-filter cut-off frequency. After DRV from AFE is applied to the sensor, the AFE must wait (PRECHARGE register should be programmed to accommodate this external RC time constant) for the sensor to settle before sampling. Sensor power-on settling time and noise-filter cut-off frequency as a function of capacitor value are tabulated in the table below.

DIFFERENTIAL CAPACITOR	SENSOR POWER-ON TIME ($2.3 * R * C$)	NOISE FILTER CUT-OFF FREQUENCY ($2\pi * R * C$) ⁻¹
0.1 nF	1.73 μs	212 kHz
1 nF	17.3 μs	21.2 kHz
33 nF	569 μs	643 Hz
0.1 μF	1.73 ms	212 Hz
0.33 μF	5.69 ms	64.3 Hz

Power-On Time & Noise Filter Cut-Off Frequencies

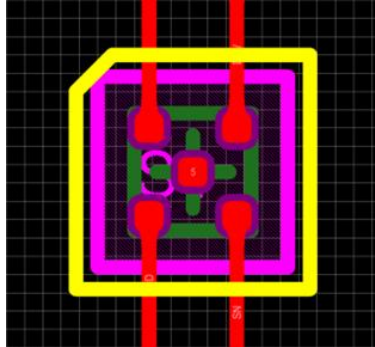
PCB Layout Best Practices

- Keep sensor and AFE on the same PCB- **REQUIRED**.
- Sensors should be designed into low mechanical/electrical/temperature stress PCB/flex areas.
- If using a flexible PCB, a top hatched ground plane is recommended.
- The MECH pin in the middle of the device should be soldered to the board for mechanical stability.
- The sensor edge should be at least 0.25 mm from all edges of the PCB – **RECOMMENDED**.
- Do not use glue/epoxy between sensor and board or around the sensor – **REQUIRED**.
- Underfill is not recommended. For harsh environments like salt spray please contact Semtech for recommendations for materials for under-fill and glob top.
- Keep clearance between all adjacent components and sensor rubber – **REQUIRED**.
- Keep epoxy-sealed components at least 3 mm away from the sensor.



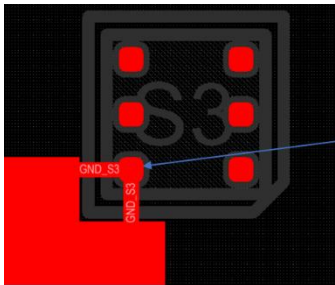
Clearance Between Epoxy Sealed Component and MEMS Force Sensor

- Route sensitive signal and power lines away from high-speed digital clock and digital interface lines.
- Symmetrical routing improves sensor self-alignment during assembly – **REQUIRED**.
- Make the traces to all the pins the same width. Semtech recommends greater than (or equal) 5 mils traces for all signals.
- Keep signal lines short (**REQUIRED**) and free of 90° turns (steep angles/corners can cause undesired acid traps during the manufacturing process). Use 45° turns or rounded-edge-turns for all signal/power lines.

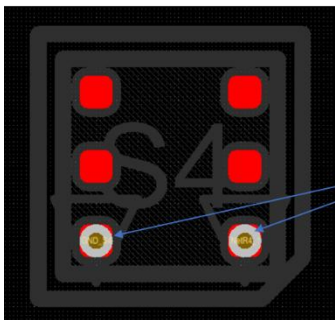


Recommended Design/Layout Practice

- For additional stability consider implementing “teardrop” routing into vias, pin pads and T-junctions; this increases the copper area and avoids steep angles in areas where multiple traces meet in a common point.
- Do not use vias directly under the sensor solder balls.



Do not use multiple traces on same pads.
Do not use copper pouring or plane direct on device pad.
This will make routing non-symmetrical with other pads.



Do not use Via in pad or micro via (Blind via) direct on land area.

NOT Recommended Design/Layout Practice

Footprint and SMT Details

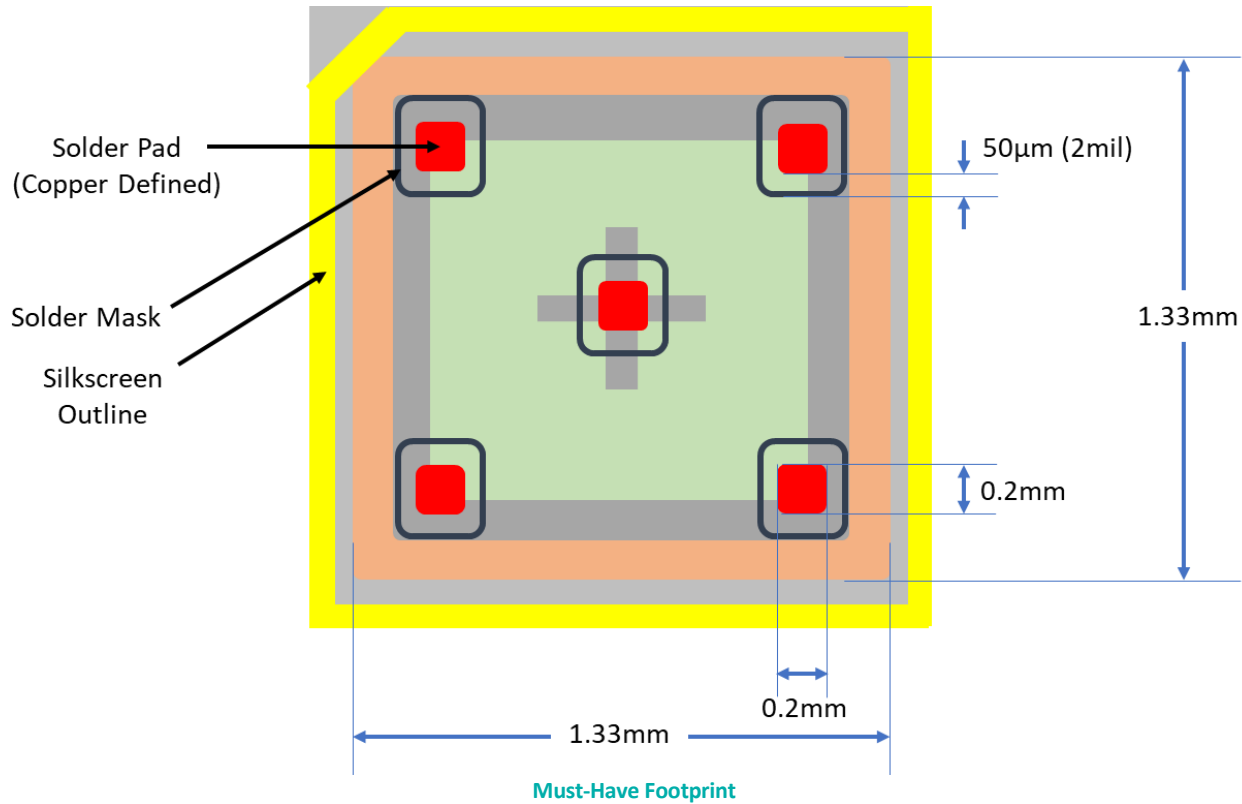


Figure above depicts the **must-have** footprint drawing for the sensor and the details below must be followed for the SMT of the sensor onto a PCB/FPCB:

- Pad dimensions and type
 - Rounded rectangle with dimensions: **7.874 mil x 7.874 mil (0.2 mm x 0.2 mm)**
 - Radius of the rounded square pad: **corner radius of 0.05 mm (not a circular pad)**
 - Copper defined, **not solder-mask defined**
- Solder mask: **2 mil (50 µm) wider on all sides of the pad.**
- Solder paste stencil thickness: **4 mil (100 µm)**
- Solder paste stencil aperture: **1:1 with the copper pad (0.2 mm x 0.2 mm)**
- Solder paste: **Alpha OM 550 HRL1 (to reduce solution offset and TCO), SAC405 or SAC305**

Device Handling – Best Practices

- Do not pick up the device with metal tweezers. Use a vacuum pickup tool.
- Do not drop MEMS devices
 - Devices are specified for up to 10,000 g shock absorption
 - 10,000 g can cause damage to the moving micromechanical structures
- Do not “snap” panelized, assembled PCBs.
 - Avoid damage to the device due to potential shock of > 10,000 g
- Do not ultrasonic clean the MEMS
 - Excessive drive motion through resonant frequency coupling can cause damage to the sensor
 - Use low-flux solder for “non-cleaning” requirements
- Follow ESD-safe handling recommendations:
 - Store sensors in ESD containers (e.g. T&R, moisture-sealed)
 - Handle devices only in ESD-safe work areas
 - People /machines handling the sensor must be grounded to avoid potential ESD damage

SMT Pick-and-Place Guidelines

Nozzle Considerations

SMT assembly machines typically employ a vacuum nozzle for Pick & Place (P&P) operations. Not all equipment is the same, but there are some general guidelines to follow, for optimal performance/yield.

- Select a vacuum nozzle only.
 - DO NOT use mechanical grippers or collets that contact the edges/sides of a WLCSP.
 - DO NOT handle WLCSP parts with metal tweezers.
- Select softer nozzle materials where possible.
 - Nozzle materials can vary (metal, ceramic, plastic, rubber), and certain equipment requires certain nozzle materials. Select softer material to avoid potential damage from mechanical shocks.
- Select an appropriate nozzle tip shape.
 - Nozzle tip shape can affect the amount of pressure applied to a single point on a WLCSP part.
 - Nozzle tips should be circular, square/rectangular, and preferably with a vertical separator.
 - The nozzle tip surface should be a single plane; do not use a nozzle with a protruded edge.



Acceptable Vs NOT acceptable Tip Shapes

- Select an appropriate nozzle tip size, for the part being placed.
 - The largest tip dimension should be smaller than the size of the WLCSP part.
 - The tip should account for the tolerance of the pick accuracy, so that it's not possible for the nozzle tip to contact the edge or the corner of a part.
 - The tip size should be as large as possible, while satisfying the above 2 points.

Pick and Place Operation

The P&P operation consists of two key steps: picking a part out of the carrier tape pocket, and placement of that part onto a circuit board. Both operations are performed with some alignment tolerance: nozzle-to-part tolerance during the pick, and part-to-board alignment during the placement. Additionally, the placement should occur with as little force as possible, to press the die into place on the board.

WLCSP parts are shipped to customers in standard carrier tape & reel packaging. Parts sit in a pocket that is slightly larger than the size of the part. A part can move laterally inside the pocket, typically by less than 0.1 mm, but this varies by product.

- If available, the P&P equipment should use a vision detection system to align the P&P nozzle to the center of the die, for every part picked.
- If a vision alignment system isn't available, the nozzle size selection should account for the positional tolerance of the WLCSP part in the carrier tape pocket.

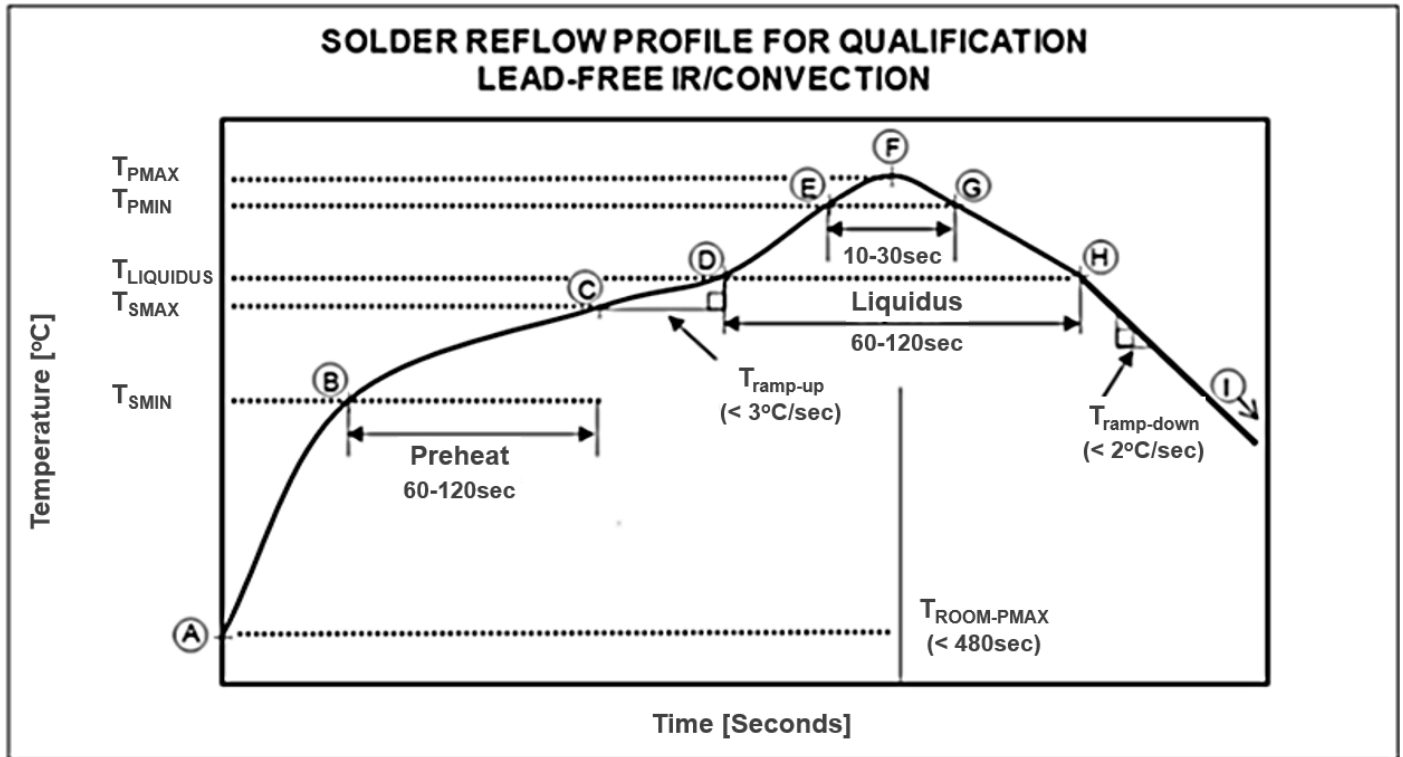
The die placement operation should be performed with a placement accuracy of ≤ 0.1 mm; placement accuracy of ≤ 0.05 mm is typical with modern P&P equipment. The placement force shall be the minimum required to bring the WLCSP solder bumps into contact with the solder paste on the board.

A contactless vacuum pickup is strongly preferred; the vacuum force alone should lift the part with only incidental surface contact to the die top.

- Use air-ejection placement in preference to contact placement. A typical air-ejection pressure is 150 mbar.
- If air placement isn't available, use a contact mode for the part placement. Ensure that both the pick and placement force is not larger than 2 N (~200 gf). Placement forces should be verified periodically using a calibrated load cell, to ensure parts aren't placed with a force over this specification.

Assembly Instructions

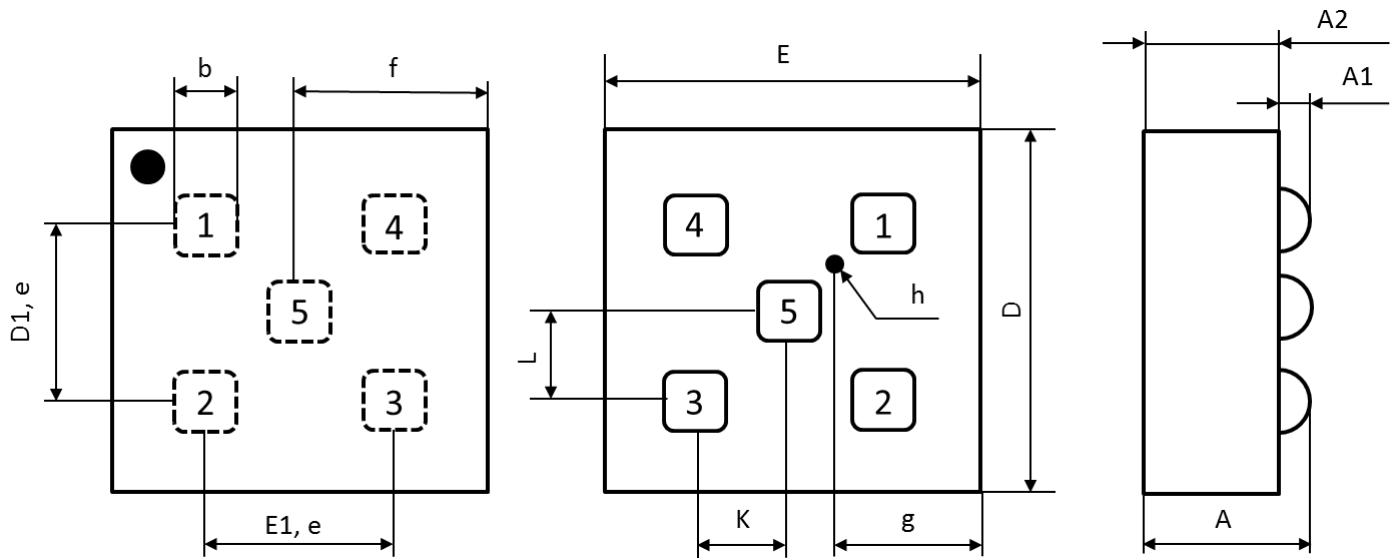
The chip is designed to be reflow-soldered using direct-chip attachment (DCA) techniques to the circuit substrate (e.g. FR-4 or FPCB) and reflow profile that is designed for RoHS and Pb-free compliance (see the profile below). Assembly houses should follow this profile closely but can choose more conservative ramp-up/down rates. To avoid damage to the chip do not exceed the maximum ratings of the qualification profile (e.g. $T_{PMAX} = 260\text{ }^{\circ}\text{C}$, measured at the top side of the WLCSP). Customers must verify with their assembly house for the appropriate reflow profile.



Soldering Profile for Reflow Assembly

STEP	PARAMETER	TEMPERATURE [$^{\circ}\text{C}$]	TIME [S]	MAXIMUM RATE [$^{\circ}\text{C/S}$]
A	T_{ROOM}	25		
B	T_{SMIN}	150		
C	T_{SMAX}	200	$60 < t_{BC} < 120$	
D	$T_{LIQUIDUS}$	217		$r(T_{PMAX} - T_{LIQUIDUS}) < 3$
E	T_{PMIN} [255 $^{\circ}\text{C}$, 260 $^{\circ}\text{C}$]	255		$r(T_{PMAX} - T_{LIQUIDUS}) < 3$
F	T_{PMAX} [260 $^{\circ}\text{C}$, 265 $^{\circ}\text{C}$]	260	$t_{AF} < 480$	$r(T_{PMAX} - T_{LIQUIDUS}) < 3$
G	T_{PMIN} [255 $^{\circ}\text{C}$, 260 $^{\circ}\text{C}$]	255	$10 < t_{EG} < 30$	$r(T_{PMAX} - T_{LIQUIDUS}) < 2$
H	$T_{LIQUIDUS}$	217	$60 < t_{DH} < 120$	
I	T_{ROOM}	25		

Mechanical Information



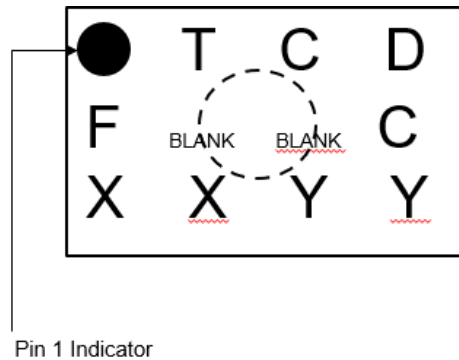
Info for 1.33 mm x 1.33 mm x 0.57 mm (Nominal)

Symbol	Min.	Nom.	Max.	Notes
A	0.544	0.569	0.595	Overall height
A1	0.064	0.069	0.075	Solder stand-off
A2	0.48	0.5	0.52	Body thickness
D	1.28	1.33	1.38	Body size
D1	0.59	0.6	0.61	Solder bump footprint
E	1.28	1.33	1.38	Body size
E1	0.59	0.6	0.61	Solder bump footprint
5				Number of solder bumps
b	0.195	0.21	0.225	Solder bump size
e	0.59	0.6	0.61	Solder bump pitch
f	0.64	0.665	0.69	Center solder bump to die edge
g	0.48	0.5	0.52	Pin 1 ID to package edge (bottom)
h	0.035	0.04	0.045	Pin 1 ID diameter (bottom)
K	0.29	0.3	0.31	Center solder bump to corner solder bump (center), X
L	0.29	0.3	0.31	Center solder bump to corner solder bump (Center), Y

All dimensions are in mm unless otherwise specified; dimensions and tolerances conform to ANSI Y14.5M-1982.

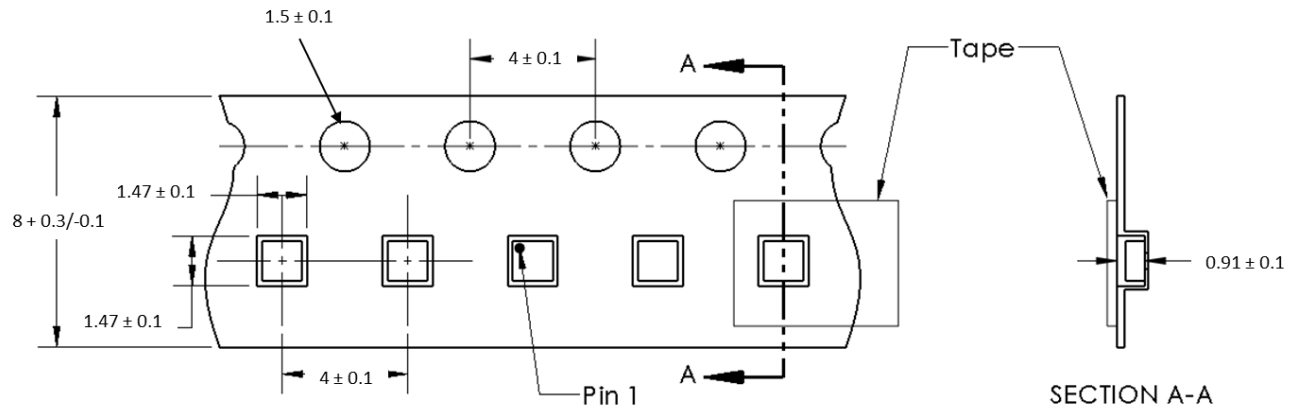
Product Marking

The sensor top mark coding consists of 3 rows of letters and numbers printed on the top of the WLCSP package.



Product Marking - Line1: Trace Code, Line2: Product ID, Line3: Die Traceability

Tape and Reel Information



Sensor Tape and Reel Dimensions

All dimensions and tolerances in this tape and reel diagram are in mm.

The image below depicts a tape reel sealed in an ESD-protective bag. The diameter of the sensor reel measures 7 inches (Radius = 3.5 inches) with reel thickness of 0.25 inches to comfortably host the tape. Each reel is stamped with the company logo, part number, lot number, and date code.



Reliability and Environmental Information

The Semtech sensor is tested and meets Level 1 (unlimited) Moisture Sensitivity Level (MSL) specifications.

Quality, Reliability and Environmental reports are available and can be furnished upon request.

GLOSSARY (Alphabetical Order)

Bridge Resistance – Resistance expressed in k Ω when measured between the positive and the negative output terminals of the sensor.

CDM (Charge Device Model) – CDM is a model for characterizing the susceptibility of an electronic device to damage from electrostatic discharge (ESD). CDM is commonly used for semiconductor devices and has been standardized in JEDEC with JESD22-C101E.

Full-Scale Range (FSR) – Full-Scale Range or FSR is the maximum operational range of the sensor and is measured in N.

HBM (Human Body Model) – HBM is the most used model for characterizing the susceptibility of an electronic device to damage from electrostatic discharge (ESD). The model simulates an ESD discharge that may occur when a human contacts an electronic device.

Hysteresis – Hysteresis is the maximum deviation, expressed as a percentage of full-scale range (%FSR), between sensor output on increasing force and sensor output on decreasing force.

Linearity – Linearity is the deviation of sensor output from a perfect linear response. Actual sensor data is “aligned” to the perfect linear response with a best straight-line fit approach. Linearity is unit-less.

Noise – Sensor noise is defined as the random deviation of the sensor’s output signal, which varies in time and is measured in nV/rt-Hz.

Sensitivity – This parameter defines how much the sensor output changes with applied force. Sensitivity is measured in mV/ N.

ZFO (Zero-Force Offset) – ZFO is the natural sensor offset in mV with no force (0 N) applied to the sensor.

Revision History

Revision	Date	Description/Changes	Pages Changed
1	04/07/2026	Initial Release	NA



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